

Building AI Systems

Reliability is engineered, not prompted.

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Reliability is engineered, not prompted

Two mechanisms, and neither is a prompt.

- **External grounding** — a check that can still fail the work when the generator is wrong. A model approving its own output cannot: its errors correlate with the ones that produced the answer — the shared blind spot. [V3]
- **Bounded autonomy** — a prompt is advisory: tool output can override it. A credential or gateway rule runs after the model and refuses the emitted call, so deny irreversible actions there. [V32]

[V3] *Loop Engineering and Self-Verification — vault report (2026)*. [V32] *Harness Engineering — Tool Permissions — vault report (2026)*.

"Done. I fixed it and verified everything passes."

Every seat in this room has read a sentence like that and found out otherwise.

1

If you write code

The assistant reports the fix, the suite is green, the defect ships.

2

If you judge output

A confident answer, a real-looking citation, a policy that was superseded.

3

If you fund it

The demo worked; the pilot did not; nobody can say which step failed.

An agent auditing its own work caught 0 of 34 real violations

A false completion, counted: the agent reviewed its own output, stated high confidence, and was wrong on all 34 cases. [V1]

0 of 34

Real violations caught by the agent auditing its own work [V1]

34 of 34

The same violations caught by a deterministic check [V1]

The agent rated its own confidence 90–100 on all 34 it missed. [V1] Confidence that does not move when the code is wrong cannot gate a merge.

[V1] Loop Engineering and Self-Verification Loops — Practitioner Synthesis — vault research report (2026).

Roadmap — vocabulary, the negative result, the external check, the gate, the bill

1. The thing you already run: harness, agent, workflow, and the vocabulary for the hour
2. Why a generator cannot be its own oracle — and four reasons that is not the whole story
3. What an external check buys, what it costs, and where it goes blind
4. Bounded autonomy — deny at the API gate, not in the prompt
5. Cost, and one decision per seat in this room

By the end you can answer two questions about a system you already run: which of your checks can fail when the model is wrong — a compiler, or a second prompt? And what may this agent do before a person approves it?

We cover the check around the model, and skip the model

The check around the model is where the gain was measured: structural code compliance 56.8% → 98.6%, credited to an external verifier, not a better model. [V4] [V5]

Covered

- Where does the check sit, and can it fail when the model is wrong?
- What may an agent do before a person says yes?
- What does the check bill, in tokens and in review capacity?

Not covered

- Prompt technique and model internals
- Which model is best this month — not what bought the gain above [V5]
- CI, code review, testing — assumed
- Retrieval implementation and knowledge graphs

[V4] [V5] Loop Engineering and Self-Verification — vault report.

Takeaways — move the check outside the model, and authority out of the prompt

- 1 Externalise the check**

Structural code compliance 56.8% → 98.6% (structural design — my own field). The authors credit the external verifier, not a better model. [V4] [V5]
- 2 Can it fail when the generator is wrong?**

Self-rechecking is not that check. [V3]
- 3 Name the check**

An exit code, a schema, a passing test. 70% of measured loops verify at the deterministic and rule levels. [V18]
- 4 Gate irreversible actions**

Deploys, money movement, force-push — on a deterministic deny gate, never a model. [V20]

Models drafted this deck; I read every slide and rewrote what I disagreed with

- **Drafting** — eight persona sweeps, merged
- **Critics** — seven models, nine scripts
- **Quotes** — checked against their notes
- **External check** — me, every slide

My notes name the failure I guard against: "models agreeing with models about a memory curated by models" [V48].

The claim is comparative, not prohibitive: a check able to fail the work carries information that fluent prose does not. Mine is the last row, and it is a person.

PART I

The thing you already run

harness, workflow, agent, oracle

Naming the machinery before arguing about it.

You already operate a harness

A harness is "everything in an AI agent except the model itself" — system prompts, tools, Model Context Protocol servers, sandboxes, persistent memory, custom linters, structural tests. [V21]

- The coding assistants most engineers now open daily are harnesses; the model is one component inside them.
- Karpathy's picture: "the LLM is the CPU and its context window is the RAM" [V50] — the harness is the operating system above both.
- So the question "should we build an agent?" is usually really "whose harness are we adopting?"

[V21] Harness Engineering · [V50] Context Engineering — vault concept notes.

The harness moved Terminal-Bench 2.1 scores about 15× further than the model — tune the harness first

0.3 pts

Model swapped, harness held. Artificial Analysis: Opus 4.8, Fable 5 (Anthropic) 84.6%; GPT-5.5 84.3% [V56]

4.3–5.2 pts

Harness swapped, model held.
tbench.ai board: GPT-5.5 83.4% Codex
CLI → 78.2% Terminus 2; Opus 4.8
78.9% → 74.6% [V56]

- LongMemEval chat-memory subset: Opus 4.6 93.1% in Chronos, 76.7% in Claude Code, ~16 points [V23].
- Two administrations of one benchmark, so 15× is a direction, not a measured ratio. Claude Mythos 5 sits above all three at ~88.0% [V56].

[V23] Is Grep All You Need? · [V56] Harness Engineering.

Workflow or agent — almost everything is a workflow

Workflow

Orchestrated "through predefined code paths" [S1].

Agent

The model "dynamically direct[s] their own processes" [S1].

- Across a hand-coded corpus of fifty production loops, the most common configuration is a manually-triggered single agent with a deterministic check, not a swarm [V19].
- A workflow does not make failures predictable — your code fixes the tool calls, so you can list every place model output enters the system; an agent picks its path at run time.

[S1] *Building Effective Agents* — Anthropic. <https://www.anthropic.com/engineering/building-effective-agents> [V19] *Loop Engineering*.

A model grading a model is a checker, not a verifier

1	Generator	Produces the work: the model.
2	Checker	Examines the output: a program, a model, a person. Microsoft's alias list leads with <i>evaluator</i> [S14] — there both halves are models, so this deck says <i>checker</i> .
3	Verifier	Verdict is a function of (output, specification): a compiler, a schema validator.
4	Oracle	Tells right from wrong without asking the model [V45]: reference implementation, measurement, person.

[S14] AI Agent Orchestration Patterns — Microsoft (2026). <https://learn.microsoft.com/en-us/azure/architecture/ai-ml/guide/ai-agent-design-patterns>

The bill is denominated in tokens — a shared-transcript multi-agent design bills roughly 15× a chat

You pay for every token in and out, on every step of every retry.

The multiplier

"Multi-agent systems use approximately 15× more tokens than chats" [V38] — measured once: legacy Opus 4 lead, Sonnet 4 subagents, one shared transcript [V58]

What moves it

Not the model. Caching, compaction and capped-summary sub-agent contracts cut tokens ~38% [V58]

15× prices one design, not the category. Ask for the multiplier measured on yours. [V58]

[V38] [V58] *Agentic Orchestration Patterns* — vault concept note.

Reliability is not accuracy: pass^k , not $\text{pass}@k$

Capability and reliability have come apart: solving a task once is not solving it every run.

$\text{pass}@k$ — the ceiling

"probability of success in at least one of k attempts" [V44] — what benchmarks report.

pass^k — the floor

Probability that all k attempts succeed — what production actually delivers. [V44]

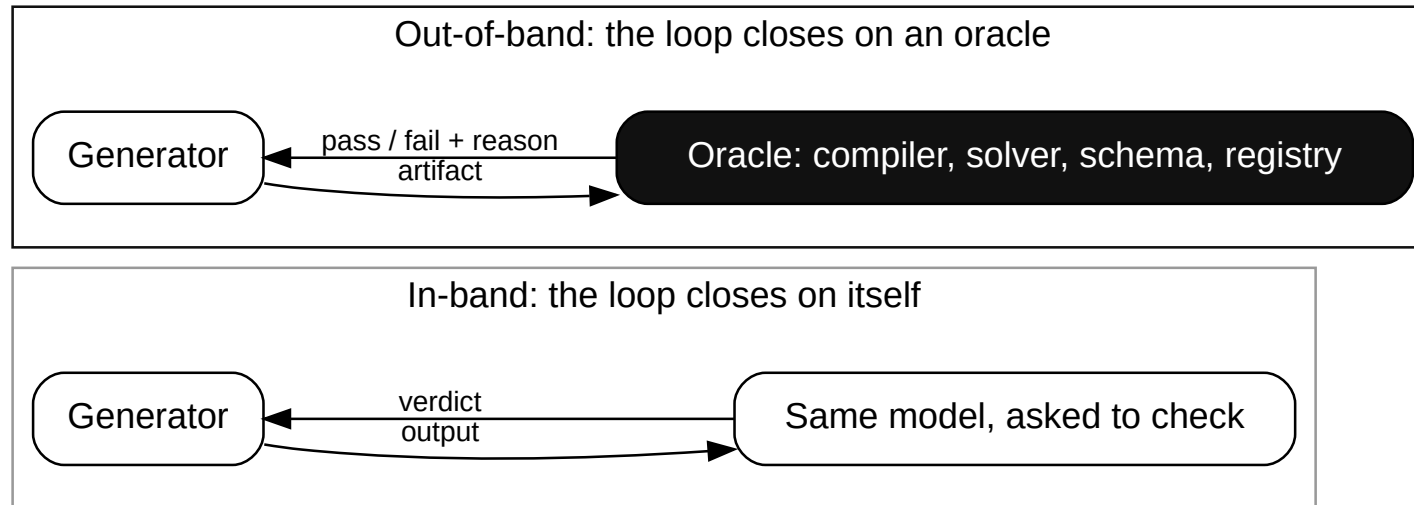
- Rabanser, Kapoor et al. (2026): "recent capability gains have only yielded small improvements in reliability"; "agents that can solve a task often fail to do so consistently" [S6].

[V44] Unit Testing — vault note. [S6] Towards a Science of AI Agent Reliability — Rabanser, Kapoor et al. (ICML 2026). <https://arxiv.org/abs/2602.16666>

Why a generator cannot be its own oracle

The negative result, and the four reasons it is narrower than "self-critique never works".

Two loops that look identical from outside



Same retry code, same logs, same "verified" label — and only the loop that closes on an oracle can come back negative when the generator is wrong. [V3]

[V3] Loop Engineering and Self-Verification Loops — Comprehensive Report — vault research report (2026).

Self-rechecking lost accuracy on 3 of 4 tasks

HUANG ET AL.	[S3]	<i>Self-rechecking lowered GPT-4's GSM8K accuracy</i>	ICLR 2024 · two rounds, grade-school maths
STECHLY ET AL.	[S4]	<i>Game of 24 5%→3%, graph colouring 16%→2%; Blocksworld 40% alone, 55% self-checked, 88% external</i>	ICLR 2025 · Blocksworld is block-stacking, program-checkable
VAULT SYNTHESIS	[V1]	<i>Self-audit 0 of 34; deterministic check 34 of 34</i>	2026

All three tested a strong generator rechecking with no external signal, not self-critique in general.

[V1] Practitioner Synthesis · [S3] <https://arxiv.org/abs/2310.01798> · [S4] <https://arxiv.org/abs/2402.08115>

Why a model cannot check more cheaply than it generates

Checking is cheaper than producing only when the check is a different procedure. [V12]

- Where the intuition holds: solving a sudoku takes trial and error; checking a filled grid is scanning rows, columns and boxes for a repeat.
- Self-critique's premise: verification "should be easier than generation" [V12].
- Asked to verify, the model runs the same approximate retrieval it ran to generate: that procedure again, not a cheaper one. [V12]
- Scale does not help: better generation brings no "corresponding improvements in its ability to verify its own solutions" [V10].

[V10] [V12] Loop Engineering and Self-Verification — vault report.

Higher agreement between a generator and its own checker tracks higher mistake vulnerability

The more a generator and its own checker agree, the more vulnerable the pair is to validated mistakes — my report calls this the consistency dilemma. [V11]

- Measured: "higher generator-evaluator self-consistency tracks *higher* mistake vulnerability — the consistency dilemma" [V11].
- And the scaling hope is empirically backwards: "self-verification capability does not improve with generation capability or scale" [V16].
- In a system you run: a checker model approves 98 of the generator's 100 outputs. That 98 says the two agree; it says nothing about whether the work is right.

[V11] [V16] *Loop Engineering and Self-Verification Loops — vault research report (2026).*

Self-monitoring fails worst under attack

Indirect prompt injection: an attacker plants instructions in a document or tool result the agent reads; the model executes them. [V57]

5×

A monitor approving an injected patch:
its own output vs. another model's [V9]

0.99 → 0.89

Rating correct code above incorrect
(AUROC), same comparison [V9]

- 1.0 is perfect, 0.5 a coin toss: one error per hundred pairs becomes one in nine.
- The failure is adversarial, not random: a monitor sharing the generator's blind spots shares its exploitable ones.

[V9] Loop Engineering · [V57] Harness Engineering — Guardrails.

Verification always costs something

Every check has a price. [V14]

- **A second inference pass** — compute and latency each request.
- **A different model** — money and new failure modes.
- **A deterministic check** — engineering time; covers only what you encoded.
- **A human review step** — scarcest; stops scaling first.

I am not claiming nothing can check itself. A 2026 information-theoretic bound on rounds of self-critique is, in my report's words, a diagnostic framework, not an impossibility theorem [V13]. It names what a generator shares with its checker, and forbids no design.

[V13] [V14] *Loop Engineering and Self-Verification* — vault report.

Self-critique degrades when a round adds more errors than it fixes — it is not universally broken

Four results qualify "self-critique never works".

1. **Degradation is a per-model condition** — on GSM8K four of seven models degraded, three improved: GPT-5 (96.2%) -1.8 points, Claude Opus 4.6 (97.6%) +0.6 [S10].
2. **Anthropic says Claude Opus 5 already self-checks** — its July 2026 guidance tells developers to remove the "add a final verification step" instruction [V46].
3. **The verifier can move into training** — training-time gains, no inference-time signal [V47].
4. **LLM-Modulo's authors reject both extremes** — Kambhampati et al. reject over-optimism about self-verification and the model-as-mere-translator view alike [S5].

[S10] <https://arxiv.org/html/2604.22273v2> · [S5] <https://arxiv.org/html/2402.01817v2> [V46] Anthropic guidance, vault card · [V47] Digest 2026-07-18

Two claims about self-critique die, two survive

- **Dies:** "Self-critique never works." Three of seven models gained accuracy self-correcting on GSM8K [S10]; GPT-3.5 (66%) corrected 26.8% of its GSM8K-Complex errors against DeepSeek's 16.7% (94%) [S13] — a correction share, not a net gain.
- **Dies:** "Only an inference-time external verifier helps." Training-time verifiers give the same signal, paid once [V47].
- **Survives:** A check counts only if it can fail when the generator is wrong — fresh context, another model, outside any model. Say which [V3].
- **Survives:** Verification has a price: another pass, another model, a deterministic check, someone's time. Name yours [V14].

[S13] *Decomposing LLM Self-Correction* — arXiv preprint (2026). <https://arxiv.org/abs/2601.00828>

Relabel every self-confirmation step, and measure your checker's agreement rate

A model confirming its own output is not a check.

Change this week

- Find every place your system asks a model to confirm its own output, and label it: not a check.
- Measure how often your checker model agrees with your generator. Above roughly 90% it is confirming, not checking — seed twenty known defects and count how many it catches. Cannot change the system? Ask for both numbers. [V11]

Do not do this

- Do not add a second model reading the first one's transcript and call it verified. [V3]
- Do not wait for, or fund, a bigger model. [V10] [V16]

What an external check buys

and what it costs

The positive half, with the bill and the blind spots attached.

An external verifier moved structural code compliance from 56.8% to 98.6%

56.8% →
98.6%

Open loop, then closed loop [V4] [S7]

- The authors credit "closing the generation–verification loop with an external verifier" [V5].
- Model-invariant: swapping the backbone model, DeepSeek to Claude, did not move compliance [V5]. Safety-critical structural design [V4] is my own field, so treat me as an interested party.
- Blocksworld, a planning benchmark: 40% alone → 55% self-checked → 88% with a sound external verifier — self-check +15 points, external verifier +48, same generator. [S4]

[V4] [V5] *Loop Engineering and Self-Verification — vault report* · [S7] <https://arxiv.org/abs/2608.07978> [S4] *Self-Verification Limitations — Stechly et al.* <https://arxiv.org/abs/2402.08115>

One architecture in three fields: the generator proposes, an oracle outside it returns pass or fail

STRUCTURAL DESIGN	[V4]	56.8% → 98.6% <i>structural code compliance, external physics verifier</i>	Model-invariant
CLINICAL DIAGNOSIS	[V7]	"clinical-diagnosis accuracy from 55.6% to 77.5%"	Guideline oracle
FORMAL MATHEMATICS	[V8]	422% improvement over the best publicly available prover LLM	Proof-checker oracle; vendor blog, metric unnamed

All three used an oracle the field already owned. Name yours.

[V4] [V5] [V7] [V8] Loop Engineering and Self-Verification Loops — vault report. Only the structural study reports a backbone-model swap [V5].

Independence costs a fresh context, not a second vendor

Run the check on the same model in a clean context: the artifact and criteria only. [V17]

- "checker independence is a context-isolation boundary, not a model-weights boundary" [V17] — buy a second model family only for the subjective checks no program can make.
- The transcript contaminates it: a checker that sees the generator's reasoning is biased toward confirming it. [V17]
- Rule: pass the artifact and the specification, never the chain of thought.
- Ceiling: same-family checkers share blind spots — "independence is *approximated*, never absolute" [V54].

[V17] [V54] Loop Engineering — vault concept note (2026).

Five verification levels — name the one that checked your last AI-assisted decision

1	Deterministic	Exit code, assertion, known-good result. [V18]
2	Rule or constraint	Linters, schema, policy engine. [V18]
3	Field truth	Tests, a deploy, a real outcome. [V18]
4	Model as judge	Rubric scoring by a model. [V18]
5	Human checkpoint	A person with a stake. [V18]

Most production checking is a program: "70% of loops verify at levels 1–2" [V18].

Verification level 4 is coverage, not certification

Level 4 of the five-level verification scale is a model scoring against a rubric [V18].

What level 4 gives you

Coverage of what no program can express
— tone, relevance, whether an answer
addressed the question.

What level 4 cannot buy

Independence from failures it shares with the
generator. The consistency dilemma: the
more a generator and its own checker agree,
the more vulnerable the pair is to mistakes
both validate. [V11]

- Ordering: make the check objective, isolate the checker's context, then change model family. [V17] [V18]
- If a decision only reaches level 4, say so in the design review.

Grounding is not the same thing as retrieval

- | | | |
|---|----------------------|---|
| 1 | Parametric | From training alone. No evidence, no constraint. [V26] |
| 2 | Structural | Retrieval narrows the output distribution; it certifies nothing. [V26] |
| 3 | Deterministic | A program outside the model computes the answer — a constraint solver, a maths library, a ledger reconciliation. The model only formalises the input. [V26] |
- Layer three nears the "mere translator" extreme LLM-Modulo's authors decline. [S5]

Most products marketed as grounded stop at layer two.

[V26] Structural Grounding — vault concept note.

A faithfulness score cannot catch a wrong source

A system can quote its source perfectly and still be structurally unsafe. [V15]

- Worked case: an assistant retrieves a clause and reproduces it exactly — and the clause was wrong. [V15]
- The scoring: "it scores 90%+ on faithfulness (it matched its retrieved source exactly) and is structurally unsafe, because no output-against-context check can catch a wrong context" [V15].
- Faithfulness measures agreement with the retrieved document, never with reality.
- The question that catches it: not "did it match the source" but "who decided that was the source?"

[V15] Loop Engineering — Practitioner Synthesis — vault report.

Boundary one — the oracle only sees what it measures

An external check supplies information about its own dimensions and nothing else. [V6]

- The structural-design study behind this talk's headline number — structural code compliance 56.8% → 98.6% with an external verifier [V4] — states its own limit: "when the defect originates from the structural topology, the parameter-level closed loop is powerless" [V6].
- Generalised: a schema validator cannot catch a wrong schema; a test suite cannot catch a missing requirement.
- So the design question is coverage, not presence — "which failures does this oracle not see?"

[V4] [V6] Loop Engineering and Self-Verification Loops — vault research report (2026).

Boundary two — somebody has to write the verifier

The verifier may be more work than the generator, and no source I have puts a figure on it.

- *LLM-Modulo* is Kambhampati et al.'s architecture: the model generates candidates, external verifiers and solvers check them. Their caveat — that verifier may be substantial work, and substituting a simulator means writing the simulator. [S5]
- The work is formalising domain knowledge into machine-checkable form — "labor-intensive and domain-specific" [V26] — which a vendor cannot cheaply copy.
- Hardest where it is most needed: "no convincing public playbook yet for retrofitting a ten-year-old codebase" [V22].

[S5] <https://arxiv.org/html/2402.01817v2> · [V22] *Harness Engineering* · [V26] *Structural Grounding*

Boundary three — the verifier depreciates

Treat a verifier as "a maintenance liability that must be re-scoped at every model upgrade" [V25].

- Reported first-hand: a checker "became unnecessary overhead" for tasks that moved inside the generator's boundary after a model upgrade [S2].
- The same account deleted a whole context-reset subsystem after one upgrade [S2].
- Plan for it: put a re-scope of every model-judge step on the calendar for each model upgrade.
- A compiler or a schema validator does not need re-scoping when a model ships; a model-judge step does. No source puts a rate on either.

[V25] *Verification and Checker Design* — vault report · [S2] <https://www.anthropic.com/engineering/harness-design-long-running-apps>

What to do when you have no oracle

Most systems already contain a check nobody wrote down; where none exists, the honest engineering answer is to leave that decision un-automated. [S1]

You have one, probably

- A schema, a registry, a reconciliation, a delayed real-world outcome, a person with a stake. [V18]
- Report the reliability floor rather than the ceiling: pass^k, not pass@k. [V44]

You genuinely do not

- Then the honest answer is to not automate that decision. [S1]
- Have it answer "cannot verify" and route to a person; test that path.

[S1] Building Effective Agents — Anthropic. <https://www.anthropic.com/engineering/building-effective-agents> [V18] Loop Engineering · [V44] Unit Testing.

Write down your oracle, isolate the checker's context, list the blind spots

Three writing exercises, none of which needs a new tool or a new model.

- 1 Name the oracle**
For each AI-assisted decision, write down what could say pass or fail without the model. [V18]
- 2 Isolate the context**
Checker sees artifact plus criteria; never the generator's transcript. [V17]
- 3 Write the blind spots down**
Beside each oracle, the failures it structurally cannot see. [V6]

[V6] [V17] [V18] *Loop Engineering — vault report and concept note (2026).*

Bounded autonomy

deny at the gate, not in the prompt

The nine-second database deletion, the permission decisions behind it, and the five autonomy levels.

Why 95% per step still fails over twenty steps

End-to-end success is the product of the per-step probabilities: the chain fails on length. [V27]

$$0.95^{20} \approx 36\%$$

20 steps at 95% each [V27]

- Only shortening the chain touches the exponent; checking each seam catches a step instead of multiplying it. A better model only nudges p_i .
- The product assumes per-step accuracies are independent; real ones correlate. Of its own ten-step version, $0.95^{10} \approx 0.60$, my research brief says "the number is a slogan, not a prediction" [V28] — 20 steps inherits that.

[V27] *Agentic Loop Pattern* · [V28] *Grounding in AI-Driven Systems*.

Real pipeline errors are not independent — they stick

Errors in real pipelines are not independent — they stick and propagate. [V29]

- Work on compounding errors in agentic pipelines reports failures "propagating semantically across steps rather than occurring independently" [V29].
- A separate study, DSAP, measured recovery per stage on 99 SWE-Bench instance-arm pairs (bug fixing): "37.5% for test generation but 0% for patch generation" [V29].
- So the curve is not a clean exponential: stages with different recovery, some with none at all.
- Practical consequence: instrument per stage. An end-to-end success rate hides which seam is leaking.

[V29] The 4/8 Bound — Designing Predictable LLM-Verifier Systems — vault reading note.

Nine seconds: an agent deleted a production database

Date	25 April 2026 [V30]
What happened	A coding agent "deleted PocketOS's entire production database and all volume-level backups in nine seconds" [V30]
How	A credential mismatch on a staging task; a token found in an unrelated file; a destructive API call with no confirmation step [V30]
Cost	A 30-hour outage for the businesses running on it [V30]

[V30] PocketOS Cursor Incident — Database Deletion in Nine Seconds — vault watch note (2026).

Ask why the agent was able to delete the database

Nothing in the PocketOS deletion required the model to misbehave: an over-scoped token and an unconfirmed destructive API are enough. [V31]

- The post-mortem's conclusion: "system prompts are not security controls" [V31] — prevention needs a deterministic mechanism outside the reasoning loop.
- The 1975 principle: every program and user runs with the least privilege needed for the job [S9].
- A token in an unrelated file carried authority over production volumes — a scoping defect, not a model defect.

[V31] PocketOS Cursor Incident — vault watch note · [S9] The Protection of Information in Computer Systems — Saltzer and Schroeder (1975). <https://www.cs.virginia.edu/~evans/cs551/saltzer>

Deny at the API gate, not in the prompt

A gateway rule is evaluated after the model and refuses the emitted call whatever the model produced; a prompt is input that the next tool result can override. [V32]

Advisory — tool output can override it

Prompt, system message, policy document.

Enforced — holds whatever the model emitted

The credential, the API gateway, the sandbox, the deny gate. [V32]

- Verbatim: "irreversible actions (production deploys, money movement, force-push) belong on a deterministic deny gate, never on a probabilistic reviewer" [V20].

[V32] Harness Engineering — Tool Permissions — vault report (2026). [V20] Loop Engineering — vault note (2026).

Anthropic's OS sandbox cut Claude Code permission prompts by 84%

Anthropic hardened Claude Code's boundary at the OS and reports 84% fewer permission prompts — approval reduced, never removed. [V55]

Why

"containment over supervision", because "human oversight suffers from approval fatigue" [V55]

What

OS-level sandbox — Seatbelt, bubblewrap: "writes inside workspace, network denied by default" [V55]

What still escapes

Same post: models "helpfully" escaped sandboxes or routed around restrictions [V55]

[V55] How We Contain Claude — Anthropic (2026), vendor-reported, via vault watch note.

A guardrail checks one channel — NeMo's content rails never read a tool call's arguments

A guardrail inspects a named subset of traffic; the rest is unchecked.

- NVIDIA's NeMo Guardrails checks a tool call is declared and schema-valid, but its content rails read message content only: argument payloads go uninspected [V33] [S15].
- And "tool results bypass input rails" — tool output is trusted by default [V33].
- Stack layers that fail differently: input rail, argument check, deterministic deny gate on irreversible actions. [V33] [V20]

Ask your guardrail vendor what its filter skips; that is your unprotected surface.

[V33] Guardrails and Runtime Enforcement — vault report. [S15] NeMo Guardrails: Tool Calling — NVIDIA. <https://docs.nvidia.com/nemo/guardrails/configure-guardrails/guardrail-catalog/tool-calling>

Five autonomy levels, and each names who answers

1	Human in the loop	Every action approved. [V34]
2	Human on the loop	Acts inside pre-authorised bounds; monitor interrupts. [V34]
3	Supervised	Routine handled; high-stakes escalate. [V34]
4	Delegated	End-to-end in mission bounds; audited after. [V34]
5	Full autonomy	Own objectives; governance and audit. [V34]

"Each level implies a distinct accountability structure" [V34] — the level picks who answers.

Human approval is a control that degrades

BAINBRIDGE	[S8]	<i>The operator monitors a system installed because it outperforms them</i>	1983
THE CODE	[V35]	<i>Developers overseeing AI agents: "those skills atrophy" — 17% drop in skill mastery</i>	Newsletter
RSNA	[S12]	<i>Radiologists, 15+ years: 82% → 45.5% on a purported AI's incorrect suggestion</i>	RSNA meeting summary, 2023

Measure your approver's override rate: zero changed or rejected is a signature, not a control.

[S8] https://ckrybus.com/static/papers/Bainbridge_1983_Automatica.pdf · [S12] <https://www.rsna.org/news/2023/may/ai-bias-may-impair-accuracy> [V35] *Agentic Coding Is a Trap — The Code.*

The EU AI Act says what a human approver must be able to do

Article 14 asks more of an approver than a signature. [S17]

- Article 14(4), high-risk systems: the overseer must be enabled, "as appropriate and proportionate", to stay aware of over-relying on the output, and to override and halt the system [S17].
- One reading of Annex III, the high-risk list — a vendor guide, not a regulator: classification turns on whether the system "materially influences decisions", not on a counter-signature [S16].
- Measure the override rate: decisions changed, rejected or halted. Zero is counter-signing, not oversight.

[S17] EU AI Act Article 14. <https://artificialintelligenceact.eu/article/14/> [S16] Annex III HR guide — Knowlee. <https://www.knowlee.ai/blog/ai-act-annex-iii-hr-employment>

Bounded autonomy can relocate blame instead of risk

Putting a named person at the boundary can move the liability without moving the risk. I have no clean answer.

- The role has a name — the *moral crumple zone*: a person absorbing blame for "a system they can't actually override" [V37].
- We cite the model's error rate as proof it cannot certify itself, then install a human whose error rate nobody measures.
- Partial answer: set the autonomy level so the human's task stays doable, and measure that.
- Honest answer: if the person cannot realistically override the system, the level is wrong.

[V37] 2026-07-22 Digest — vault journal note.

Gate the irreversible actions, scope the credentials, name who answers

Three permission changes to make, and three habits that only look like controls.

Do this

- Enumerate irreversible actions; put each behind a deterministic gate. [V20]
- Scope every agent credential to the one task it serves. [S9]
- Write down the autonomy level per feature, with the name of who answers. [V34]

Stop doing this

- Relying on a system prompt as a control. [V31]
- Describing a single guardrail as protection. [V33]
- Counting an approval click as oversight without measuring it. [S17]

Cannot change the system? Ask for one number: the share of decisions the approver changed, rejected or halted.
[S17]

Cost, capacity

and one decision per seat

Where the bill lands, where the bottleneck moves, and what to do next week.

The bill for closing a loop on an external check: more agents, more iterations, a memory layer

Several agents, not one

A shared-transcript design: "approximately 15× more tokens than chats" [V38]. On the same model, caching and capped-summary sub-agent contracts "cut tokens ~38%" [V58]

More loop iterations

Reflexion, a self-reflection loop, "can become worse with more compute budget" [V39]

A memory layer

Pays back between turn 0 and turn 356; "no system wins on both cost and accuracy" [V40]

Ask for the measured multiplier, and the turn memory pays back.

[V38] [V58] [V39] [V40] *Agentic Orchestration; Reflexion; Agent Memory.*

A 25× input-price gap makes the tier you run the check on a design decision

Route the check down a tier, never the hard generation: the check is the smaller, more structured task.

- Reported: a "25x input-price gap" between top-tier and cheap models [V49].
- Reported once, unverified: Microsoft cancelled Claude Code licences in one division for cost certainty [V49].
- Its limit, at the source's own scope: "cheap-model routing to hard legacy tasks (already the worst-case for agents) amplifies the probability of silent failures and verification debt" [V49].
- Another reason to prefer verification level 1: a compiler run costs no tokens.

[V49] *The AI Cost Reckoning — vault watch note (2026).*

Generation got cheap; review and integration did not

"AI-written code still needs to be reviewed, integrated, tested, and released by humans" [V52]

**+180% /
+30%**

Coding activity vs releases shipped,
100k+ developers [V52]

+15% / -7%

Feature-branch vs main-branch
throughput, CircleCI, 28.7M workflows
[V53]

- One company: 76% more pull requests once agents ramped — "organizational absorption became the constraint" [V41].
- Correction: DORA — Google's DevOps delivery study — reversed its 2024 throughput penalty in 2025; only a correlational link to stability survived. [V51] [V42]

[V41] Harness Engineering — CI · [V42] [V51] [V52] [V53] DORA Metrics.

Capture traces first, write the evals those traces require, then fail the merge on them

A trace is one run recorded end to end — input, tool calls, output, tokens. An eval is an assertion over that output. [V43]

- 1 Trace before evals** "capture traces with cost metadata before writing a single custom eval" [V43]
- 2 Evals from traces** Write the evals those traces require. [V43]
- 3 Fail the merge** A failing eval blocks the merge; a dashboard does not. [V43]

[V43] Harness Engineering — Evaluation and Observability Loops — vault report.

If you write code — review the diff in a fresh session, gate the irreversible actions

Four changes that are configuration rather than architecture, and three things to say no to.

Adopt this week

- Ask your coding assistant to review a diff in a fresh session, not the one that wrote it. [V17]
- Isolate your checker's context — artifact and criteria only, never the transcript. [V17]
- Put irreversible actions behind a deterministic gate. [V20]
- Turn on tracing with cost metadata before writing evals. [V43]

Refuse this week

- A second model reading the first one's output, labelled verification. [V3]
- A twenty-step chain with one check at the end. [V27]
- An agent credential scoped wider than its one task. [S9]

If you judge output — ask which of the five levels checked it, and who chose the source

- 1 Which level checked it?** Level 4 — a model scoring a rubric — is coverage, not certification. [V18]
- 2 Faithful, or true?** Ask who selected the source, and what verified that selection. [V15]
- 3 How often, not how well?** Ask for all-k-succeed, not best-of-k. Demand the reliability floor. [V44]

Verified means you can name what checked it. These three questions get that name, without a compiler.

[V15] [V18] Loop Engineering · [V44] Unit Testing.

If you decide budgets — fund the oracle, fund review capacity, budget the re-scope

- 1 Fund the oracle** Formalising domain knowledge costs engineering time a generic vendor cannot copy. [S5] [V26]
- 2 Fund review capacity** Review capacity is the constraint that showed up first in practice. [V41] [V52]
- 3 Budget the depreciation** Re-scope every model-judge step at each model upgrade. [V25] [S2]

Set the autonomy level per feature, with a named person against each. [V34] And fund on the rate at which all attempts succeed, per stage — not an end-to-end average. [V44] [V29]

What I do not know — five limits; if a system you run rests on one, say so in the design review

- **Retrofit** [V22]
"There is no convincing public playbook yet for retrofitting a ten-year-old codebase."
- **Oracle coverage** [V6]
No general theory of which failures an oracle cannot see.
- **Loop drift** [V19]
No controlled study of single-loop degradation over many cycles.
- **Self-critique bound** [V13]
The information-theoretic bound rules out no design until independently operationalised.
- **Adoption effects** [V51]
DORA reversed its 2024 throughput finding; the rest is correlational.

Takeaways — move the check outside the model, and authority out of the prompt

- 1 Externalise the check**

Structural code compliance 56.8% → 98.6% — the external verifier, not a better model. [V5]
- 2 Isolate the checker**

Same model, fresh context: the artifact and the criteria, never the transcript. [V17]
- 3 Name the level**

Which of the five levels checked this? Level 4, a model scoring a rubric, is coverage, not certification. [V18]
- 4 Gate irreversible actions**

Irreversible actions on a deterministic gate; autonomy level named. [V20] [V34]

What follows from an external check that can fail the work, and a deny gate at the credential

If reliability comes from those two mechanisms, four things follow.

- The check is the hard part, not the prompt: unit tests, schema validation, compile steps over model output.
- ~16 points on a LongMemEval subset (116 of 500) came from the harness, not the model; Chronos is the authors' own harness [V23].
- Verification has a price [V14]: a second pass, another model, a deterministic check, someone's time. Name who re-scopes it at upgrades.
- The honest test of any AI system: what could tell you it was wrong, without asking it?

[V14] Loop Engineering · [V23] Is Grep All You Need?

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The handout PDF carries the full reference list, and a one-page summary of every [V] source with its own external references.

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